

Youtu-GraphRAG: Vertically Unified Agents for Graph Retrieval-Augmented Complex Reasoning

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Outline

- 1 Introduction
- 2 Youtu-GraphRAG
- 3 Experiments

GraphRAG and Youtu-GraphRAG

- GrapgRAG: improve either graph construction or graph retrieval in isolation.
- Youtu-GraphRAG: jointly connect the entire framework as an intricate integration.
 - Schema-Bounded Agentic Extraction.
 - Upon Schema: Graph Indexing with Knowledge Tree.
 - Schema-enhanced Agentic Retriever.



Schema-Bounded Agentic Extraction

- Schema:

$$\mathcal{S} \triangleq \langle \mathcal{S}_e, \mathcal{S}_r, \mathcal{S}_{\text{attr}} \rangle,$$

- Extract triples:

$$\mathcal{T}(d) = \{ (h, r, t), (e, r_{\text{attr}}, e_{\text{attr}}) \mid \{f(h), f(t), f(e)\} \in \mathcal{S}_e, \{r, r_{\text{attr}}\} \in \mathcal{S}_r, e_{\text{attr}} \in \mathcal{S}_{\text{attr}} \}.$$

- Adaptive design: dynamically refines the initial schema $\mathcal{S}$ through continuous interaction with the document content.



Upon schema: Graph index

1.Entity Representation

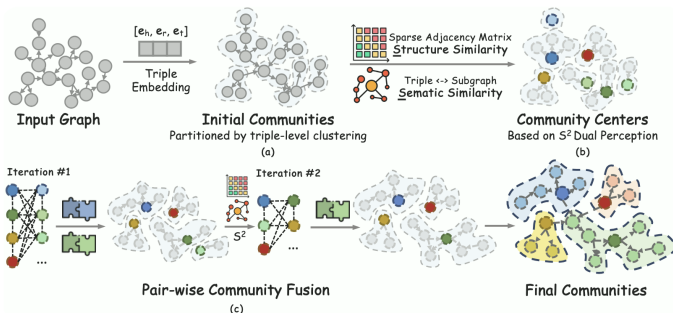
$$\mathbf{e}_i = \frac{1}{|\mathcal{N}_i|} \sum_{(e_i, r, e_j) \in \mathcal{N}_i} [\mathbf{e}_i \parallel \mathbf{r}_{ij} \parallel \mathbf{e}_j].$$

2.Cluster Initialization

- applying K-means clustering on the entity embeddings $C_1^0, \dots, C_k^0$.

3.Iterative Community Fusion via Dual-Perception Scoring

$$\phi(e_i, C_m) = \underbrace{S_r(e_i, C_m)}_{\text{relational}} \oplus \lambda \underbrace{S_s(e_i, C_m)}_{\text{semantic}},$$



- Upon Schema-bounded extraction: raw graphs transforms into structure Knowledge Tree K.

$$L_\ell = \begin{cases} \{\mathcal{C}_m\} & \ell = 4 \text{ (Community)} \\ \{\arg \max \phi(v_i, \mathcal{C}_m)\} & \ell = 3 \text{ (Keywords)} \\ \{(h, r, t) \mid h, t \in \mathcal{E}, r \in \mathcal{R}\} & \ell = 2 \text{ (Entity-Relation Triples)} \\ \{(e, \text{has_attr}, \{e_{\text{attr}}^{\text{type}} : e_{\text{attr}}^{\text{value}}\})\} & \ell = 1 \text{ (Attributes)} \end{cases}$$

Agentic Retriever

- Schema-enhanced Query Decomposer:

$$\mathcal{Q} = f_{\text{LLM}}(q, \mathcal{S}) = \{q_1, q_2 \dots q_i\}.$$

- Iterative Reasoning and Reflection:

$$\mathcal{A}^{(t)} = \underbrace{f_{\text{LLM}}(q^t, \mathcal{H}^{(t-1)})}_{\text{Reasoning}}, \underbrace{\mathcal{H}^{(t-1)}}_{\text{Reflection}}$$

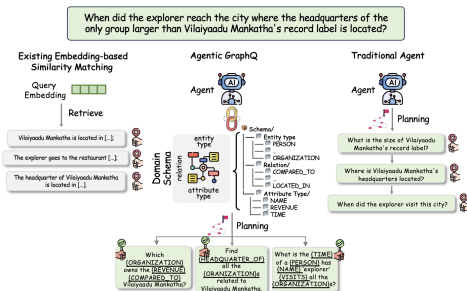
- Multi-Route Retrieval:

$$\text{Entity Matching : } \arg \max_{e \in \mathcal{E}} \cos(e, q_i)$$

$$\text{Triple Matching : } \arg \max_{(h,r,t) \in \mathcal{G}} \cos((e_h, r, e_t), q_i)$$

$$\text{Community Filtering : } \arg \max_{c_n \in \mathcal{C}} \cos(c_n, q_i)$$

$$\text{DFS Path Traversal : } \mathcal{P}(q_i) = e_0 \xrightarrow{r_1} e_1 \xrightarrow{r_2} \dots \xrightarrow{r_n} e_n \quad \text{s.t.} \quad \forall r_i \in \mathcal{R}, n \leq d$$



Result on benchmark

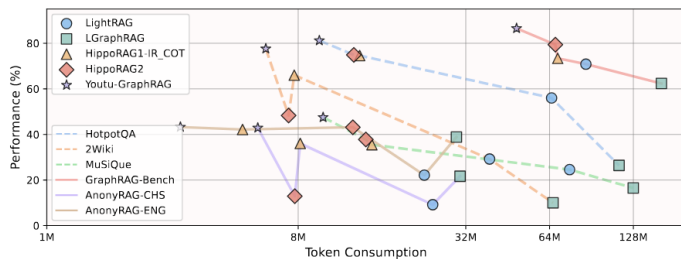
- dual-mode evaluation: Open mode; Reject mode.
- anonymous datasets: AnonyRAG-CHS, AnonyRAG-ENG.

CHS	Raw Query	从史进拜师到最终离开少华山，期间共经历了哪些身份转变？
	Annoy Query	从[PERSON#277]拜师到最终离开[LOCATION#759]，期间共经历了哪些身份转变？

Result on benchmark

Method	HotpotQA		2Wiki		MuSiQue		G-Bench	Annoy-CHS	Annoy-ENG
	Open	Reject	Open	Reject	Open	Reject	Open	Open	Open
Deepseek-V3-0324									
Zero-shot LLM	53.70	-	41.6	-	25.7	-	70.92	9.62	8.18
Naive RAG	79.90	72.40	70.3	38.9	47.49	30.63	71.81	12.5	43.02
E ² GraphRAG	68.70	48.80	43.20	20.00	28.36	8.01	68.66	16.01	35.97
RAPTOR	80.90	73.60	70.10	38.40	48.50	31.10	73.08	12.08	40.2
LightRAG	71.90	56.00	58.00	29.20	38.98	24.57	70.83	9.16	22.14
GraphRAG	56.10	26.40	41.80	10.00	32.20	16.50	75.54	21.66	38.85
G-Retriever	49.00	6.70	35.80	5.00	23.50	1.70	70.63	4.07	5.08
HippoRAG	81.70	73.10	77.90	64.00	48.30	36.20	72.89	36.77	40.68
HippoRAG-IRCOT	81.00	74.60	78.40	66.00	46.70	35.50	73.38	36.05	42.17
HippoRAG2	81.80	74.90	77.30	48.30	50.80	37.80	79.37	12.92	43.16
Ours w/o Agent	83.70	75.30	72.80	57.80	51.40	40.00	81.53	37.06	40.05
Youtu-GraphRAG	86.50	81.20	85.50	77.60	53.60	47.50	86.54	42.88	43.26
Qwen3-32B									
Zero-shot LLM	36.40	-	33.30	-	13.40	-	70.04	5.11	6.49
Naive RAG	75.00	69.00	58.50	39.60	40.64	33.03	72.69	7.56	26.84
RAPTOR	79.20	72.90	61.20	40.10	38.99	32.86	72.20	13.37	22.14
HippoRAG	77.00	71.80	72.80	62.50	40.60	32.10	75.64	8.58	32.30
HippoRAG-IRCOT	80.30	76.60	74.80	65.40	44.70	37.40	77.11	9.16	33.15
HippoRAG2	81.80	71.30	65.20	39.90	51.40	37.70	80.35	12.65	38.36
Ours w/o Agent	83.80	73.90	74.90	55.30	52.90	40.10	80.74	34.88	35.13
Youtu-GraphRAG	85.90	78.60	85.70	74.20	54.60	45.30	84.48	39.24	40.05

Token Consumption



Ablation study

Variants	HotptQA	2Wiki	MuSiQue	G-Bench	AnonyRAG-CHS	AnonyRAG-ENG
w/o Comm.	79.50	75.10	44.00	85.02	39.97	39.92
w/o Agent	75.30	57.80	40.00	81.53	37.60	40.05
w/o Schema	77.10	73.40	45.60	83.50	35.61	40.32
Youtu-GraphRAG	81.20	77.60	47.50	86.54	42.88	43.26